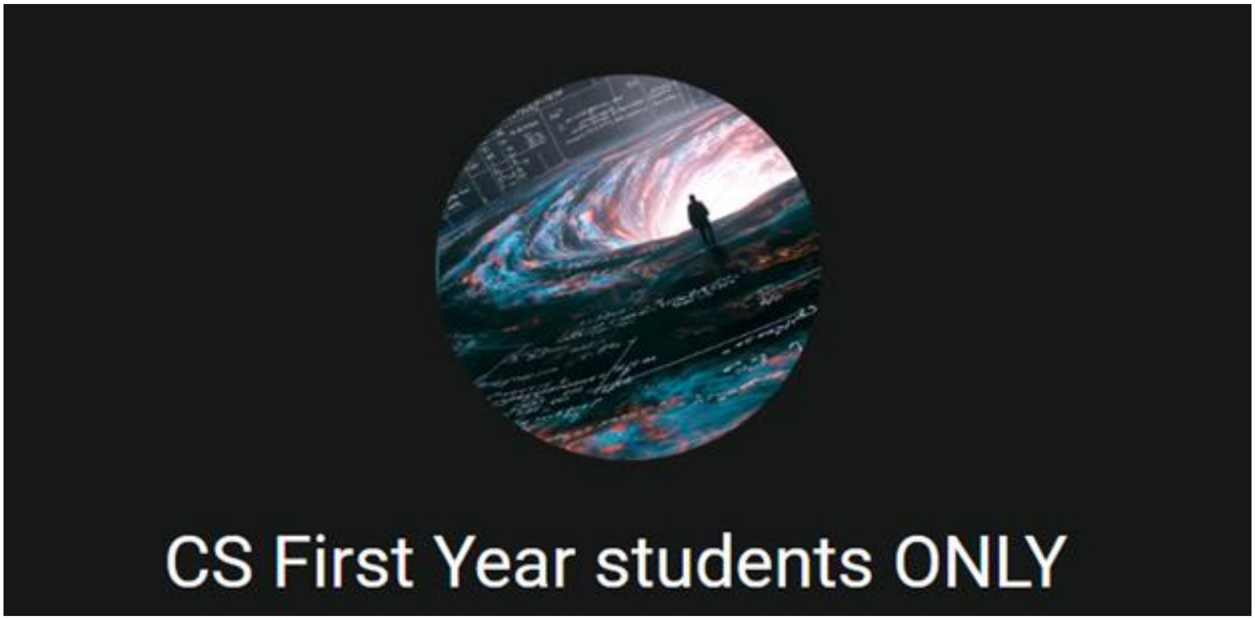
Final Exam For Second Semester (2024/2025)



Leader's :

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Ministry of High Education

Obour High Institute for Management & Informatics

معاهد العبور
Obour Institutes

Final Exam for Second Semester (2024/2025)

Course Name: Statistics and Probabilities	Course Code: BS 110 Bylaw: 2024
Year: First Model (C)	Program: Computer Sciences
Date of Exam: Tuesday 27/5/2025	Time Allowed: 2 hours Total Points: 60
Course Instructor: Assist. Prof. Saeed Hemeda	No. of questions: (3) No. of pages: (2)
Dean Approval:	

Exam instructions

- Make sure that the question form is identical to the answer sheet form
- The answer sheet must be in good condition from the moment received until it is delivered after taking the exam
- Do not bend the answer sheet or lean on it while answering

Answer the following questions:

Question One: [20 points] (ILOs: a1, a3, a4, b1, b2, c1,c2, d1)

Choose the correct answer from (A), (B), (C) or (D), to complete the sentence:

1. If $P(A) = 0.2$, $P(B) = 0.5$ and $P(A \cap B) = 0.1$ then $P(A \mid B) =$

(A) 0.50	(B) 5.0	(C) 1	(D) 0.2

❖ Given the data set {11, 2, 9, 4, 6, 3, 8}:

2. Median =…….

(A) 6	(B) 4.5	(C) 5.5	(D) 11.23

3. The interquartile range=……..

(A) 3.87	(B) 4.5	(C) 1.67	(D) 6

4. The probability of a leap year selected at random contain 53 Sunday is:

(A) 2/7	(B) 1/7	(C) 53/ 366	(D) 53/365

❖ The following table shows the height (cm) and weight (kg) for 10 students:

Height (X)	160	165	171	156	166	172	167	159	162	174
Weight (Y)	65	60	68	61	65	70	66	61	66	75

5. $\overline{X}$=…… $\frac{\text{مجموع الارقام}}{\text{عددهم}}$

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(A) 140.2	(B) 165.2	(C) 120.3	(D) 177.5

6. $\overline{Y}$=……

(A) 65.7	(B) 55.7	(C) 45.7	(D) 0

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7. r_{XY}=......

(A) 0	(B) -0.8	(C) 0.8	(D) 1

8. The correlation between X and Y is ...... correlation.

(A) perfect	(B) moderate	(C) directly	(D) inversely

9. Mean deviation of the data set { 7, 9, 7, 8, 5, 5, 6, 9}=...... $\frac{1}{8}\sum|X-\bar{X}|=1.25$

(A) 0.25	(B) 1	(C) 1.25	(D)) 2.25

10. If $p(A) = 0.2, p(B) = 0.4, p(A \cup B) = 0.6$ then $p(A \cap B)$=......

(A) 0	(B) 0.3	(C) 1.2	(D) -0.4

Question Two: [20 points] (ILOs: a1, a2, b1, b4, c2, c3, d1)

For the following paired data, calculate the required measures:

Height (cm)	160	165	171	156	166	172	167	159	162	174
Weight (Kg)	65	60	68	61	65	70	66	61	66	75

1. [10 points] Harmonic mean of weight (HM). Q_1 Q_2 $\frac{10}{\frac{1}{65}+\frac{1}{60}+\cdots+\frac{1}{75}}=63.47$
2. [10 points] Correlation coefficient between height and weight (CC). $r=\frac{Cov(X,Y)}{S_x S_y}=0.47$

Question Three: [20 points] (ILOs: a1, a3, b1, b2,b3,c1, c3, d2)

1. [10 points] Let the data set {5,5,5,5,7,7,1,1,2,3,3,4,4,4,4}; calculate the kurtosis coefficient.
2. [10 points] Given the data in the following table:

Interval	50-	60-	70-	80-	90-	100-	110-120
Frequency	7	9	14	19	14	6	2

Calculate Semi Inter quartile Range (SIR).

End of Questions
Good Luck,,,

Examiners:
Assist. Prof. Saeed Hemeda
Assist. Prof. Hassan Algendy